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Response Time of Wetland Biodiversity to Road Construction on Adjacent Lands

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Abstract: *Road construction may result in significant loss of biodiversity at both local and regional scales due to restricted movement between populations, increased mortality, habitat fragmentation and edge effects, invasion by exotic species, or increased human access to wildlife habitats, all of which are expected to increase local extinction rates or decrease local recolonization rates. Species loss is unlikely to occur immediately, however. Rather, populations of susceptible species are expected to decline gradually after road construction, with local extinction occurring sometime later. We document lags in wetland biodiversity loss in response to road construction by fitting regression models that express species richness of different taxa (birds, mammals, plants, and herptiles) as a function of both current and historical road densities on adjacent lands. The proportion of variation in herptile and bird richness explained by road densities increased significantly when past densities were substituted for more current densities in multiple regression models. Moreover, for vascular plants, birds, and herptiles, there were significant negative effects of historical road densities when the most current densities were controlled statistically. Our results provide evidence that the full effects of road construction on wetland biodiversity may be undetectable in some taxa for decades. Such lags in response to changes in anthropogenic stress have important implications for land-use planning and environmental impact assessment.*

Tiempo de Respuesta de la Biodiversidad de Humedales a la Construcción de Caminos en Tierras Adyacentes

Resumen: *La construcción de caminos puede resultar en significativas pérdidas de biodiversidad tanto a escala local como regional debido a la restricción de movimiento entre poblaciones, incremento de la mortalidad, fragmentación de hábitat y efectos de borde, invasión de especies exóticas o mayor acceso de humanos a hábitats silvestres, con lo cual se espera que se incrementen las tasas locales de extinción o disminuyan las tasas locales de recolonización. Sin embargo, es improbable que la pérdida de especies ocurra inmediatamente. Más bien, se espera que las poblaciones de especies susceptibles declinen gradualmente después de la construcción del camino, extinguiéndose localmente poco tiempo después. Documentamos la pérdida de biodiversidad en humedales como respuesta a la construcción de caminos ajustando modelos de regresión que expresan la riqueza de especies de diferentes taxa (aves, mamíferos, plantas, reptiles y anfibios) como una función de las densidades actual e histórica de los caminos en tierras adyacentes. La proporción de variación en anfibios, reptiles y aves incrementó significativamente cuando las densidades históricas fueron sustituidas por densidades actuales en los modelos de regresión múltiple. Más aun, hubo efectos negativos significativos de las densidades de caminos históricas para plantas vasculares, aves, anfibios y reptiles cuando las densidades actuales fueron estadísticamente controladas. Nuestros resultados proporcionan evidencia de que los efectos de la construcción de caminos sobre la biodiversidad de humedales pueden se indetectables para algunos taxa por décadas. Tales rezagos en la respuesta a cambios en el estrés antropogénico tienen implicaciones importantes en la planificación de uso del suelo y la evaluación de impacto ambiental.*

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Introduction

Road construction and the associated increase in human presence represents one of the major ways humans transform landscapes. Of particular concern are the effects of roads on biodiversity. Evidence is accumulating that road construction may result in significant loss of biodiversity at both local and regional scales due to restricted movement of species between local populations (Merriam et al. 1989; Baur & Baur 1990; Reh & Seitz 1990; Yanes et al. 1995; Vos & Chardon 1998), increased mortality (van Gelder 1973; Goodman et al., 1994; Rosen & Lowe 1994; Fahrig et al. 1995; Fowle 1996), habitat fragmentation and edge effects (Mader 1984; Andrews 1990; Soulé et al. 1992; Thiollay 1993; Henjum et al. 1994; Wigley & Roberts 1994), invasion by exotic species (Cowie & Warner 1993; Lonsdale & Lane 1994), or increased human access to wildlife habitats (Clark et al 1993; Young 1994), all of which are expected to increase local extinction rates or decrease local recolonization rates.

Road construction has been shown to have negative effects on a variety of ecosystem attributes, including biodiversity. For land-use planners and decisionmakers, however, the real question is, for a given set of construction options (size of roadway, type of surfacing, siting, traffic volume, etc.), what are the ecological risks and expected socioeconomic benefits? In standard economic analysis, "benefits"—or more precisely, the estimated monetary value of these benefits—are calculated with explicit reference to some time horizon: for example, short-term benefits such as employment during road construction versus long-term benefits such as increased access to natural resources, increased land values and hence, tax revenues. Similarly, any assessment of the ecological effects of road construction should include both short and long time horizons. Yet to the limited extent that environmental impact assessments provide quantitative predictions of expected ecological effects (Treweek 1996), these are invariably short-term predictions or (perhaps worse still) long-term predictions based on extrapolation from short-term assessments.

Clearly, the time dimension is important in assessment of the effects of roads or any other stressor: if ecosystem attributes show lagged responses to changes in stress levels, or if the full response takes some time to manifest itself, then ecosystem responses will be underestimated if the assessment interval—the time between the change in stressor levels and the assessment of ecosystem response to this change—is too short. If ecosystem response (or lack thereof) is determined by means of statistical hypothesis testing, the chances of detecting a (true) response to changes in stressor levels decreases as the assessment interval decreases (i.e., the Type II error rate increases, regardless of the experimental design). Thus, one may conclude there is no detectable effect of road construction when in fact an effect would have been detected had we waited longer. More generally, an accurate assessment of the full ecological effects of roads depends on knowing the time scale over which ecosystems respond to the stresses arising from road construction.

In a previous study (Findlay & Houlahan 1997), we found that for a sample of wetlands in southeastern Ontario, current herptile, bird, and vascular plant species richness showed significant negative correlations with the (more or less contemporaneous) density of roads on adjacent lands, up to distances of at least 2 km from the wetland. From this and ancillary evidence, we inferred a causal relationship, namely, that road construction leads to reduced species richness because of one or more processes (e.g., restricted movement of species). Our analysis, however, did not include the issue of temporal scale. In particular, we noted that if road construction leads to local extinction, it is unlikely to do so immediately. Rather, populations of susceptible species are expected to decline gradually, with complete loss occurring sometime later. If this is true, then the relationship between species richness and road densities estimated from roughly contemporaneous estimates of road density and wetland biodiversity may underestimate the long-term effects of road construction (Findlay & Houlahan 1997).

Methods

Because there is likely to be a lag between road construction and the onset of local extinctions, we were led to investigate the effects of changes in road densities over the past half century on current species richness. Our approach was based on the following simple model of an ecosystem (in this instance, a wetland) responding to changes in the level of some stressor (e.g., the density of roads on adjacent lands). Following road construction, the ecosystem response may not begin until after a time interval (lag L) has elapsed. Moreover, a further interval (the equilibration time τ) may elapse before the system equilibrates (if it does) with respect to the new stress levels (Fig. 1). The lag and equilibration time determine the system memory (M), the time in the past beyond which previous changes in stress levels have no appreciable effect on the current ecosystem state: $M = L + \tau$.

For environmental impact assessment, knowledge of τ and L is crucial. For example, if the effect of a change in stress X at time t is assessed at some time t', with $t < t' < t + L$, no effect will be detected, perhaps leading to the conclusion that the change in X has had no effect. Conversely, assessments at any time t', with $t < t' < t + M$ may underestimate the true magnitude of the effect (Fig. 1). Perhaps most importantly if $M > 0$, then the ecosystem will continue to change even after X stabilizes.

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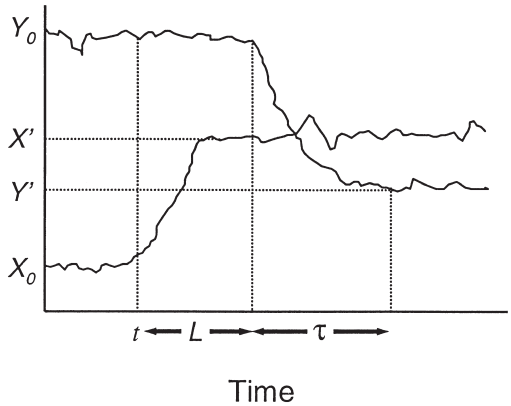

Figure 1. Hypothetical response of an ecosystem to a change in stress levels. A change in the stress level from X_0 *to* X' *fails to elicit a response until after a lag* L. *The equilibration time* τ *is the time required for the system to move from its initial response level* Y_0 *to the new equilibrium response level* Y', *assuming that the system equilibrates. In this hypothetical case, the assessment of ecosystem response at a time* t', *with* $t < t' < t + L$, *leads to the conclusion that the change in stress levels from* X_0 *to* X' *has no effect on* Y, *whereas an assessment at time* t', *with* $t + L < t' < t + L + \tau$, *underestimates the effect of the change.*

How do we estimate M? If, for a given system, time series for both stresses X and ecosystem responses Y were available, some information might be gleaned by investigating the behavior of the cross-correlation function or the cross-spectrum between $Y(t)$ and $X(t - j\Delta t)$ for different lags $j\Delta t$, $j = 0, 1, \ldots$, with Δt the interval between successive measurements of X and Y. Unfortunately, such data rarely exist for most systems of interest. A more common situation is when, for a sample of qualitatively similar ecosystems (e.g., wetlands, lakes, forests in a given geographical area), the current (time T) level of Y is known for each ecosystem, but levels of X are known only for time T and other (infrequent) times prior to T. Suppose that in this sample of ecosystems, the average memory is $\overline{M}$. Then the absolute value of the (possibly nonlinear) correlation r_{RS} between $Y(T)$ and $X(t)$ increases as $|T - t| \rightarrow \overline{M}$ (Fig. 2). Hence, given a sequence $X(T)$, $X(t_1)$, $X(t_2)$,. . . extending back in time from the present T, a rough estimate of $\overline{M}$ is given by $T - t^*$, where t^* is the time t in the past when $|r_{Y(T)X(t)}|$ is a maximum (Fig. 2).

We assessed lags by means of regression models relating species richness of various wetland taxa $S(T)$ to road densities $R(t_j)$ at different times t_j in the past. Two different procedures were used. In the case in which $\overline{M}$ is small, so that current richness depends only on changes in the recent past, once contemporaneous road densities $R(T)$ were controlled for, little of the remaining variability in current species richness $S(T)$ was explained by road densities in the more distant past $R(t_j)$. But if $\overline{M} > T - t_j$ once the effects of $R(t_i)$ for all $T > t_i > t_j$ were accounted for, a significant amount of the remaining variability in $S(T)$ would be explained by the addition of an $R(t_j)$ term to the model. In contrast, if $\overline{M} < T - t_j$, then addition of an $R(t_j)$ term to the model would not improve model fit because we would gone beyond the system memory. Thus, by proceeding sequentially through the $R(t_j)$s, we established rough upper and lower bounds for $\overline{M}$.

If road densities at successive time periods are autocorrelated, then the associated regression model will include correlated predictors. This multicollinearity means that standard errors of model parameters will be inflated and the resulting tests of null hypotheses will be correspondingly conservative. In other words statistical detection of the effect of the historical road densities will be difficult if contemporaneous densities are included in the model. Although this procedure can establish rough upper and lower bounds for $\overline{M}$, it cannot provide insight into the relative importance of changes in stress levels within the system memory. For example, although current wetland species richness may reflect cumulative changes in road density over, say, the last 20 years, the largest effects may be due to changes in the past 3 years. Alternatively, perhaps the biodiversity die was cast 10 years ago, in which case subsequent changes in road densities will have little effect. We distinguished between these possibilities by substituting historical road densities for current road densities in regression models. For example, if current species richness were caused mainly by changes in road densities occurring in the last 3 years, then a model that substitutes road densities from 10 years ago for road densities 3 years ago would provide a poorer fit. In contrast, if the die was indeed cast 10 years ago, then the former would give the better fit.

Using Government of Canada 1:50,000 topographic maps, we determined historical road densities on lands adjacent to the sample of wetlands described in Findlay and Houlahan (1997). Historical road densities were obtained for three different time periods, corresponding to the year of the aerial photographic surveys on which the maps were based. These periods were [mean ± 1 SD (year)]: 1982 ± 1.2 (t_1), 1968 ± 4.7 (t_2), and 1944 ± 7.6 (t_3). Because aerial photographs were not taken in the same year for all wetlands in the sample, there is considerable variability among wetlands in the dates of (and hence, intervals between) successive time periods. For each time period, we estimated road densities for four overlapping distance intervals from the wetland edge (0–250, 0–500, 0–1000, and 0–2000 m) by measuring the total length of all roads (both paved and loose-surface) and dividing by the area (obtained by planimetry) of each distance interval. Although it is likely that paved and unpaved roads have different ecological effects, for

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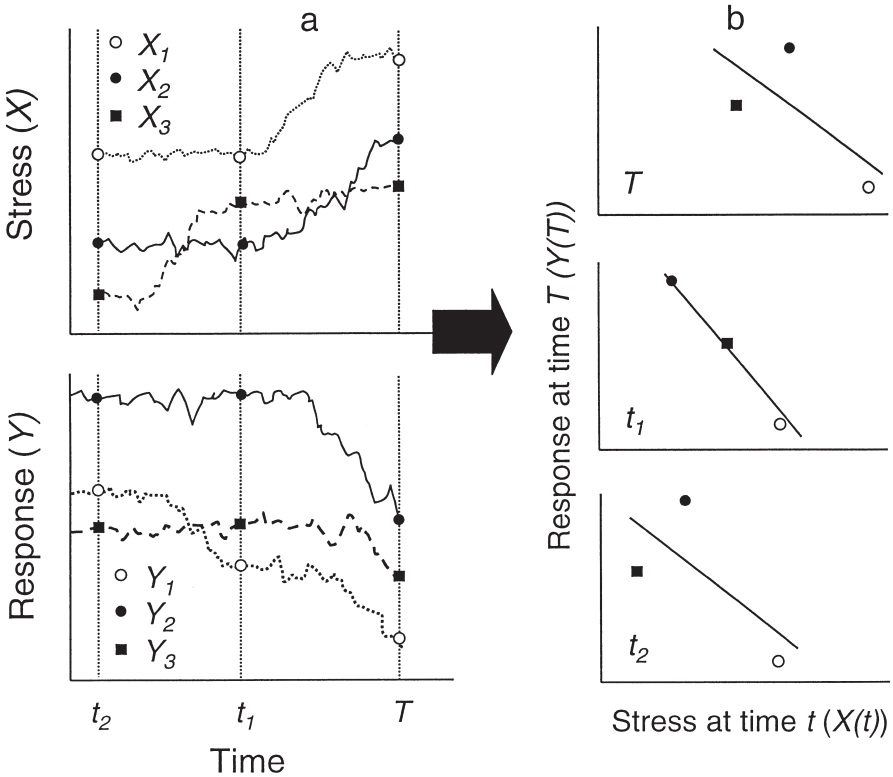

Figure 2. Effect of time lags on the detection of ecosystem response to stress: (a) changes in stress levels X_i *and ecosystem responses* Y_i *in three different ecosystems (surveys are conducted at times* t_2, t_1 *and* T *[the present]); (b) the relationship between ecosystem response at time* T *and stress based on stress levels measured contemporaneously with ecosystem response (*T*) and at different times in the past (*t_1, t_2*). The statistical detection of ecosystem response to stress depends on the lag. For no lag (*T*) and long lag (*t_2*), the null hypothesis of no relationship between ecosystem response and stress tends to be accepted. In contrast, with an intermediate lag (*t_2*) there is a strong negative relationship, suggesting that the average system memory* $\overline{M}$ *satisfies* $0 < \overline{M} < T - t_2$.

the purposes of our analysis we combined them, because for all times other than the most recent, many wetlands had no paved roads in their immediate vicinity.

For birds and vascular plants, a base model that included a constant and the logarithm of wetland area was first fit because previous results (Findlay & Houlahan 1997) showed a significant area effect for all taxa. For herptiles the base model also included the amount of forest cover within 2 km of the wetland (Findlay & Houlahan 1997). Subsequently, for each taxon, a model consisting of the base model and road density at the most recent time t_1 ($R(t_1)$) was fit for each distance interval. Examination of the historical effects of road density was then restricted to that distance interval for which the additional variance in species richness explained by $R(t_1)$ was the largest. For herptiles and plants this interval was 2 km from the wetland; for birds it was 1 km. Then, for each taxon, five more models were fit, two in which increasingly historical road densities (t_2, t_3) were substituted for the most recent (t_1) values and three in which pairs of historical road densities were included.

Results

Historical Changes in Road Densities

The time periods we considered span about four decades. During this time, the density of paved roads on lands adjacent to the wetlands tripled (repeated-measures analysis of variance, $F_{2,58} = 20.9, p < 0.001$ within 1 km; $F_{2,58} = 34.5$, $p < 0.001$ within 2 km), whereas the density of loose-surface roads declined by about 40% ($F_{2,58} = 5.59, p = 0.006$ within 1 km; $F_{2,58} = 8.94, p < 0.001$ within 2 km) (Fig. 3). This pattern was due primarily to the paving of what had been loose-surface roads. In only a few cases were paved roads created de novo, and these corresponded to the construction of major highways or primary carriageways. Total (paved and loose-surface) road density within 1 km of wetlands showed no significant increase over the period ($F_{2,58} = 0.98$, $p = 0.37$), whereas total density within 2 km showed a marginally significant increase ($F_{2,58} = 2.79, p = 0.07$). In the latter instance, virtually all the change was accounted for by increases between 1969 and 1982.

The pattern of historical change is made somewhat clearer by considering changes in the proportion of wetlands in the sample without any loose-surface or paved roads within 1 or 2 km (Fig. 4). All wetlands in the sample had loose-surface roads within 1 km even at the earliest time period. The proportion (within 1 km) without paved roads decreased dramatically over time, from 63% in 1944 to 20% in 1982. Within 2 km, 43% had no paved roads at the earliest time frame, and 0% had no paved roads by 1982.

Wetland Biodiversity and Road Density

We examined the effects of total road densities measured at different times in the past on the current spe-

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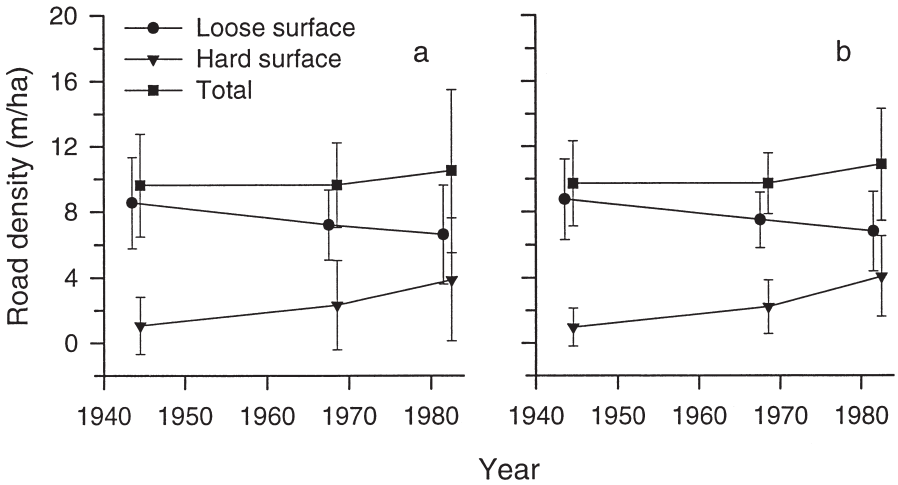

Figure 3. Average (±1 SD) road densities on lands within (a) 1 and (b) 2 km of sampled wetlands in southeastern Ontario. Times correspond to the year of the aerial photo (averaged over wetlands in the sample) from which road densities were estimated.

cies richness of vascular plants, birds, and herptiles. For all three taxa, there was evidence of significant historical effects. For both plants (Table 1) and herptiles (Table 2), there was a significant negative effect of road densities at time t_3 when the effects of the most recent road densities were statistically controlled. For plants, no significant effect of road density was detected at t_1 or t_2; this effect was evident only at time t_3 (Table 1). Furthermore, for both plants and herptiles, model fit increased as increasingly historical road densities were substituted for the most recent values (Tables 1 & 2, Fig. 5). For birds, there was evidence of a strong negative effect associated with the most recent road densities, an effect that diminished as older values were substituted. Yet when wetland area and the most recent road densities were statistically controlled, there was still a marginally

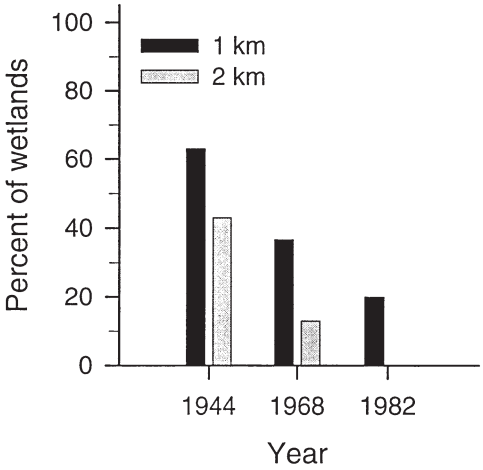

Figure 4. The proportion of wetlands with no paved roads within 1 and 2 km at different times in the past. Times correspond to the year of the aerial photo (averaged over wetlands in the sample) from which road densities were estimated and are identical for each road type.

significant negative effect of road densities at time t_3 (Table 1). For both birds and herptiles, models that included road densities at both t_1 and t_3 resulted in an additional 10% and 8%, respectively, of the variability in species richness being explained, compared to models that included only one time period (Tables 1 & 2). For plants, the increase in explained variability was negligible (Table 1).

One potential confounding factor is the serial autocorrelation of road densities at successive time periods. Not surprisingly, road densities at successive time periods were more highly correlated ($r(t_1, t_2) = 0.59, 0.72$; $r(t_2, t_3) = 0.44, 0.62$ for 1 and 2 km, respectively) than were those at t_1 and t_3 ($r(t_1, t_3) = 0.26, 0.35$). The relatively high correlation between t_1 and t_2 is the principal reason that, for all taxa, statistically significant effects were not detected for either $R(t_1)$ or $R(t_2)$ when both were included in a regression model (Table 1 & 2). In contrast, the relatively low correlation between t_1 and t_3 allowed for the detection of an effect of the latter when the former was included in fitted models.

Discussion

Findlay and Houlahan (1997) detected significant negative relationships between the richness of herptiles, birds, and vascular plants in a sample of wetlands from southeastern Ontario and the most current road densities on adjacent lands. We showed that for these same wetlands, the proportion of variation in herptile and bird richness explained by road densities on adjacent lands increased significantly when past densities were substituted for more current densities in multiple-regression models. Moreover, for vascular plants, birds, and herptiles, there were significant negative effects of historical road densities when the most recent densities were statistically controlled. For wetland plants, our results suggest that in fact, species loss may not even be detectable

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Table 1. Parameter estimates, coefficient of determination (r^2), and residual mean square (*RMS*) for multiple-regression models expressing current wetland species richness of birds and vascular plants as a function of wetland area and road densities on adjacent lands.

*Taxon/model**	β_A	β_i	β_j	r^2, *RMS*
Birds				
$M_0 + R(t_1)$	0.18 ± 0.061 (0.007)	−0.022 ± 0.007 (0.004)		0.42, 0.031
$M_0 + R(t_2)$	0.22 ± 0.062 (0.002)	−0.026 ± 0.011 (0.028)		0.40, 0.036
$M_0 + R(t_3)$	0.22 ± 0.062 (0.002)	−0.018 ± 0.015 (0.240)		0.32, 0.041
$M_0 + R(t_1) + R(t_2)$	0.18 ± 0.062 (0.007)	−0.024 ± 0.008 (0.008)	−0.008 ± 0.016 (0.608)	0.50, 0.032
$M_0 + R(t_2) + R(t_3)$	0.21 ± 0.065 (0.003)	−0.005 ± 0.015 (0.741)	−0.024 ± 0.012 (0.068)	0.42, 0.032
$M_0 + R(t_1) + R(t_3)$	0.21 ± 0.060 (0.002)	−0.022 ± 0.010 (0.036)	−0.024 ± 0.013 (0.073)	0.51, 0.032
Vascular plants				
$M_0 + R(t_1)$	0.21 ± 0.043 (<0.001)	−0.006 ± 0.007 (0.371)		0.49,0.016
$M_0 + R(t_2)$	0.21 ± 0.042 (<0.001)	−0.015 ± 0.013 (0.262)		0.50, 0.016
$M_0 + R(t_3)$	0.21 ± 0.038 (<0.001)	−0.023 ± 0.008 (0.009)		0.59, 0.012
$M_0 + R(t_1) + R(t_2)$	0.21 ± 0.044 (<0.001)	−0.002 ± 0.009 (0.792)	−0.011 ± 0.016 (0.474)	0.50, 0.016
$M_0 + R(t_2) + R(t_3)$	0.21 ± 0.038 (<0.001)	−0.002 ± 0.013 (0.871)	−0.024 ± 0.009 (0.019)	0.59, 0.013
$M_0 + R(t_1) + R(t_3)$	0.21 ± 0.039 (<0.001)	−0.004 ± 0.006 (0.561)	−0.023 ± 0.008 (0.013)	0.60, 0.013

**Models are of the form* $\log S = a + \beta_A \log A + \beta_i R(t_i) = M_o + R(t_i)$ *and* $\log S = M_o + \beta_i R(t_i) + \beta_j R(t_j)$, *where* $\log S$ *and* $\log A$ *are the* $\log_{10}$ *of wetland species richness and wetland area respectively, and* $R(t_i)$ *is the density of roads (in m/ha) within 1 km (birds) or 2 km (plants) of the wetland at time* t_i. *Cell entries are parameter estimates ±1 SE, with the associated Type I two-tailed probabilities in parentheses.*

until several decades after road construction. For birds and herptiles, in contrast, species loss is detectable within < 8 years on average. But even for these taxa, our results suggest that the full effects of road construction on species richness may not be evident for decades.

The causal mechanisms underlying the observed stress-response relationships are largely unknown. For birds and herptiles, increased mortality through roadkill is expected to reduce population growth rates. There is ample evidence that, especially for these groups, roadkill can be substantial (e.g., Goodman et al. 1994; Rosen & Lowe 1994; Fahrig et al. 1995; Evink et al. 1996; Fowle 1996). If population sizes are initially reasonably high, their reduction to the point where a species fails to be recorded in a wetland survey may take several decades to occur. Consequently, relationships between species richness—the measurement endpoint selected in this study—and road density are unlikely to be detected until this period has elapsed. The same sorts of considerations apply if the observed relationships reflect the effects of habitat loss or degradation (e.g., Mader 1984; Andrews 1990; Soulé et al. 1992; Thiollay 1993; Wigley & Roberts 1994) or—as might apply especially to plants—invasion by exotic species (Cowie & Warner 1993; Lonsdale & Lane 1994). Again, such effects are unlikely to occur instantaneously but rather will accumulate over time, so that some time must elapse before effects on species richness are detected.

Our study adds to the increasing body of evidence that populations (Doak 1995; Lignon & Stacey 1996), communities (Milchunas & Lauenroth 1996), and ecosystems (Kirkman et al. 1996; Chapin & Starfield 1997; Shao et al. 1998) may exhibit substantial lags and inertia in response to stress. These lags may reflect bona fide environmental or ecological factors; statistical, sampling, or experimental design factors; or some combination thereof. For ex-

Table 2. Parameter estimates, coefficient of determination (r^2), and residual mean square (*RMS*) for multiple-regression models expressing current herptile richness as a function of wetland area, forest cover, and road densities on adjacent lands.

*Model**	β_A	β_F	β_i	β_j	r^2, *RMS*
$M_0 + R(t_1)$	0.24 ± 0.057 (<0.001)	0.55 ± 0.151 (0.001)	−0.019 ± 0.009 (0.051)		0.61, 0.027
$M_0 + R(t_2)$	0.25 ± 0.051 (<0.001)	0.58 ± 0.139 (<0.001)	−0.040 ± 0.015 (0.013)		0.62, 0.026
$M_0 + R(t_3)$	0.26 ± 0.045 (<0.001)	0.42 ± 0.131 (0.003)	−0.041 ± 0.010 (<0.001)		0.71, 0.020
$M_0 + R(t_1) + R(t_2)$	0.24 ± 0.053 (<0.001)	0.59 ± 0.142 (<0.001)	−0.009 ± 0.011 (0.389)	−0.030 ± 0.019 (0.134)	0.68, 0.023
$M_0 + R(t_2) + R(t_3)$	0.26 ± 0.045 (<0.001)	0.45 ± 0.133 (0.002)	−0.016 ± 0.016 (0.311)	−0.033 ± 0.012 (0.010)	0.75, 0.018
$M_0 + R(t_1) + R(t_3)$	0.24 ± 0.044 (0.006)	0.47 ± 0.125 (0.001)	−0.014 ± 0.007 (0.055)	−0.036 ± 0.010 (0.001)	0.79, 0.015

**Models are of the form* $\log S = a + \beta_A \log A + \beta_F F + \beta_i R(t_i) = M_0 + R(t_i)$ *and* $\log S = M_0 + \beta_i R(t_i) + \beta_j R(t_j)$, *where* $\log S$ *and* $\log A$ *are the* $\log_{10}$ *of wetland herptile richness and wetland area respectively,* F *is the proportion of current forest cover within 2 km of the wetland, and* $R(t_i)$ *is the density of roads within 2 km of the wetland at time* t_i. *Cell entries are the same as in Table 1.*

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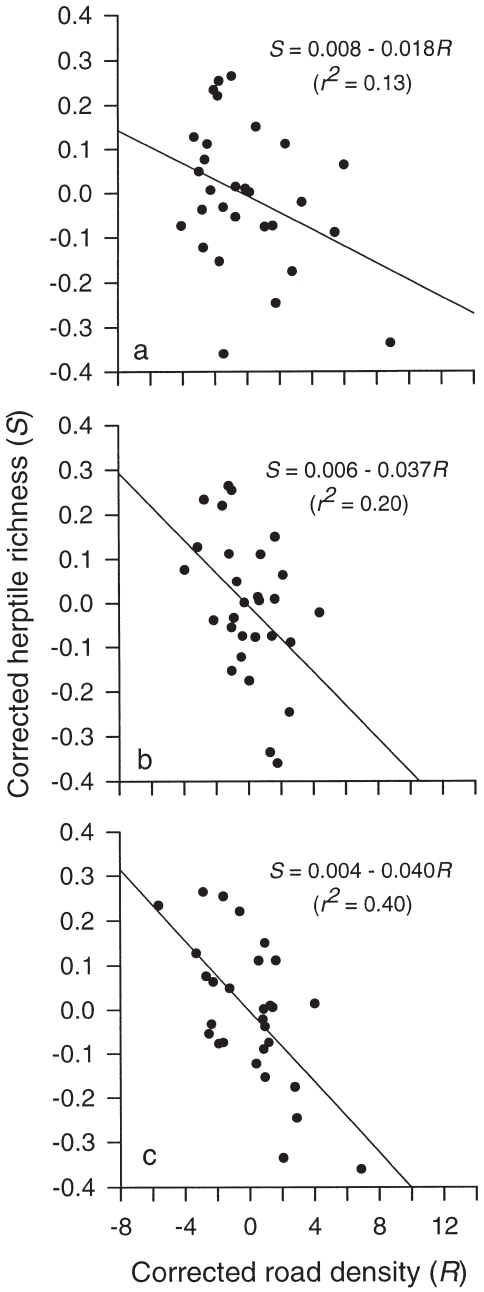

*Figure 5. Partial plots of herptile richness (*S*) versus road densities (*R*) on adjacent lands within 2 km at progressively historical times* t_1 *(a),* t_2 *(b), and* t_3 *(c). Corrected herptile richness and road densities are the residuals of a multiple-regression model expressing* S *(*R*) as a function of wetland area and forest cover within 2 km at time* t_1*, respectively.*

ample, if the selected ecological attribute (measurement endpoint) shows, even under unstressed conditions, a low spatial or temporal signal-to-noise ratio, the statistical detection of ecosystem response may not be possible until considerable time has elapsed, unless sample sizes are large (an unlikely situation, especially if the endpoint is an ecosystem-level attribute). This will be the case even when the attribute in question begins to respond immediately to changes in stress levels. Thus, as has been emphasized by a number of authors (e.g., Cormack 1994; Underwood 1994; Findlay & Zheng 1997; Stow et al. 1998), the determination of the appropriate temporal (and spatial) scales for ecological monitoring and impact assessment is crucial if the results are to be of much value.

Alternatively, observed lags may reflect, at least in part, the nature of the endpoint itself. The surveyed presence or absence of a species depends on its population size: as population size is reduced, the likelihood of it being recorded as present decreases, whereas the likelihood of it being recorded as absent increases. Consequently, changes in species richness will necessarily show lagged responses to changes in the factors responsible for reductions in population density (e.g., Gaston et al. 1998). If road construction on adjacent lands reduces population size, then at some point the population may shrink below the level detectable in a wetland survey (i.e., the species is "absent"). And because species richness is an attribute that depends on the population dynamics of all species in the wetland, it will change only at discrete intervals, corresponding to those points when a species previously present is now absent. Hence, simply by the nature of the endpoint and the ecological processes that determine it, we expect wetland species richness to show a lagged response to road construction (Fig. 6).

In assessment of the effects of human activities, a key issue concerns the sensitivity of particular ecological attributes. All other things being equal, we would tend to select attributes that show rapid and predictable responses to anthropogenic stress. For some stressors in some systems, metrics based on community-level attributes (e.g., species richness) show rapid responses to changes in the level of anthropogenic stress. For example, Karr et al. (1987) showed that an index of biotic integrity (IBI) based on the richness of different fish taxa showed a marked decline in the first year (and a subsequent slow recovery) following upstream channel restoration that resulted in downstream sedimentation. In contrast, our results indicate that considerable time may elapse between road construction and local extinction of wetland populations, leading to reductions in species richness. Consequently, our results suggest that total species richness is not a particularly sensitive indicator of the effects of road construction on wetlands, at least those in southeastern Ontario.

Our results have at least three important implications. First, they suggest that short-term environmental assessments of road construction on measures of wetland biodiversity such as species richness are likely to substantially underestimate the real effects. As such, accurate estimates can be obtained only when historical and present effects are integrated in cumulative impact assessment (Harris

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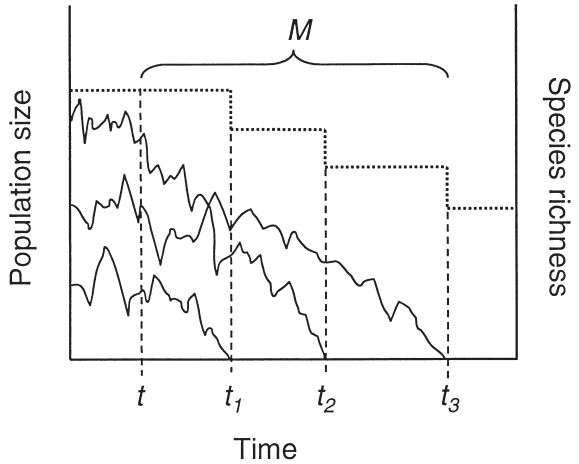

Figure 6. Hypothetical response of wetland species richness to road construction. A road is built at time t, *and individual populations of susceptible species begin to decline soon thereafter. Changes in species richness are observed only when population sizes of individual species are reduced to the point where the species in question cannot be detected in a wetland survey. Even under continuous monitoring, these changes will be observed only at discrete times: one species will be lost by* t_1, *another by* t_2, *and so on. If reductions of fewer than three species are undetectable given the nature of the monitoring regime (i.e., three represents the minimal detectable difference), species richness will show an observed lag of length* M *in response to road construc-*

1988; Johnston 1994; McCold & Saulsbury 1996). Second, as was the case for present road densities (Findlay & Houlahan 1997), the negative effects of historical road densities were detected on adjacent lands up to 1 or 2 km from the wetland. This provides additional support for the contention (Findlay & Houlahan 1997) that current Canadian provincial and federal wetland policies are inadequate insofar as the designated buffer zones, where road construction is prohibited, extend at most several hundred meters from the wetland's edge. Third, and perhaps most distressing, our results suggest that even if no new roads are constructed, wetland biodiversity will likely continue to decline in lagged response to historical increases in road densities. In a sense, the current picture of wetland biodiversity in southern Ontario—gloomy though it is—is nonetheless rosier than it ought to be.

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